YEAR 11 MATHS METHODS INVESTIGATIVE TASK 2

Probability

Prerequisites

- Basic understanding of combinatorics
- ${}^{n}C_{r}$ notation
- The multiplication and addition principle for combinatorics
- Algebra skills
- Basic spreadsheet skills

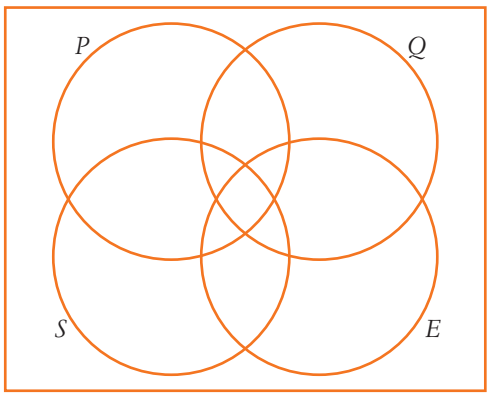
Calculators, including Computer Algebra System (CAS) calculators may be used.

[Total: 100 marks]

1 Consider the universal set to be positive integers from 1 to 20 inclusive.

- Let P be the set of prime numbers.
- Let Q be the set of multiples of three.
- Let E be the set of even numbers.
- Let S be the set of numbers that are less than nine.

a Place each of the positive integers from 1 to 20 on the Venn diagram below.



b List the elements of the sets below.

i $P \cap E =$

ii $Q \cap S =$

iii $S \cap \overline{P \cup E \cup Q} =$

iv $P \cap S \cap Q =$

v $\overline{P \cup E} =$

vi $\overline{P \cup Q} =$

c If the universal set consisted of all positive integers, show how to find the smallest number which won't fit into any of these sets.

d Use set notation to identify any null set. If the universal set consisted of all positive integers, would this still be a null set? Explain why or why not.

e Use set notation to identify any set with cardinality 1. If the universal set consisted of all positive integers, would this still have cardinality 1? Explain why or why not.

f If an icosahedral (20 sided) die numbered 1 to 20 was rolled, what is the probability of rolling:

i a prime number less than 9?

ii an even number which is not a prime number?

iii a number that is either even or less than 9, but not prime?

g Write the three events in part **f** in set notation.

[5 + 6 + 1 + 2 + 2 + 3 + 3 = 22 marks]

2 The 450 students of Year 10 and Year 11 at a fairly isolated school were asked if they wanted to start at 8 a.m. in the morning rather than at 9 a.m. Many were vehemently against the proposal and the results of the survey are tabled below.

Against the proposal	370
Male students against the proposal	210
Year 10s against the proposal	234
Year 11s against the proposal	136
Boys for the proposal	30
Female year 10s for the proposal	10

In Year 11 there were 77 girls and 123 boys and in year 10 there are 133 girls and 117 boys.

a Summarise the results in the table below

	Boys		Girls		
	For	Against	For	Against	Totals
Year 10					
Year 11					

b Use the table to find the probabilities below if one person is selected at random from the group of students who were surveyed. Express your answer as a fraction and a percentage rounded to the nearest whole number.

i $P(\text{Male})$

ii $P(\text{For the proposal})$

iii $P(\text{For the proposal} \mid \text{Year 10})$

iv $P(\text{For the proposal and a girl})$

v $P(\text{For the proposal and a Year 11 girl})$

vi $P(\text{For the proposal and a girl} \mid \text{Year 11})$

vii $P(\text{A girl} \mid \text{for the proposal})$

viii $P(\text{Year 11} \mid \text{a boy against the proposal})$

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c Determine whether or not the events 'Year 10' and 'For the proposal' are independent.

[5 + 8 +5 = 18 marks]

3 Ben and Ann celebrate their engagement by going out to dinner.

- There are two choices of entrée – prawns or mushroom and asparagus.
- There are three choices of main course – barramundi, steak or beef pie.
- There are two choices of dessert – date pudding or berries with cream.

Ben and Ann are so excited that they order three courses at random.

a How many different meals could Ben choose?

b Find the probability that Ben orders steak and berries with cream.

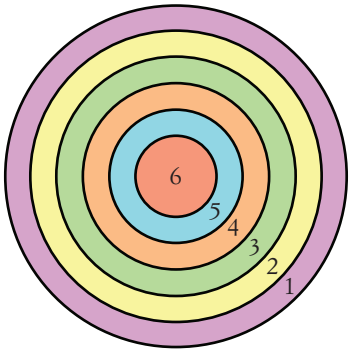
c Find the probability that Ann and Ben order different dishes for every course.

d Ann remembers she cannot eat seafood. How many meals can she choose?

e Given **d**, what is the probability Ann ordered steak and berries with cream?

[1 + 2 + 5 + 2 + 1 = 11 marks]

4 Young Jimmy makes his own dart board. Each circle's radius increases by the same length of the inner circle's radius.



If a dart hits the board, then the player gets the score that is in that ring. For example, a dart in the yellow section of the board gets a score of 2. If you completely miss, you may keep trying until you hit one of the numbers.

a With three different throws of a dart, in how many different ways can the player score a total of 6? (The order matters.) List them.

b In two throws, what is the maximum score possible?

c Jimmy constructs a table giving the probability of obtaining each score. Calculate each probability as a fraction and insert the decimal equivalent in the table, rounded to two decimal places.

Score	*P*(score)
6	
5	
4	
3	
2	
1	

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d Use a spreadsheet to check your answer to part **c**.

e Assuming that a player hits the board at any random position with three consecutive darts, find the exact (fraction) probabilities for:

i $P(2 \text{ then } 5 \text{ then } 6)$

ii $P(6 \cap 6 \cap 6)$

iii $P(\overline{6 \cap 6 \cap 6})$

iv $P(\text{a total of } 6)$.

f Look at the pattern of the probabilities in part **c**.

i Would this be any different if the dartboard was oval, but the same pattern was used to calculate the oval's radiuses? What about a square dartboard?

ii List the probabilities for a similar dartboard with 10 circles.

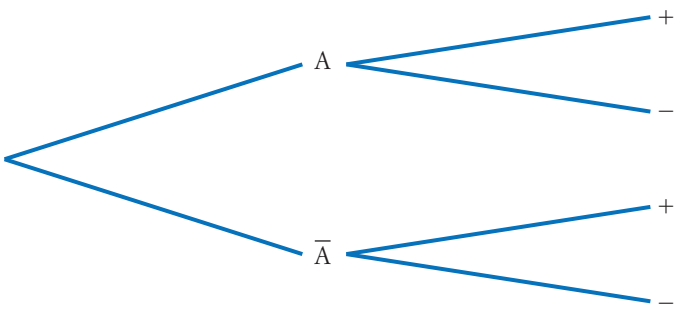
iii What would the probabilities be for a board with n circles.

iv What does this tell you about the sum of the first n odd numbers?

[2 + 1 + 8 + 5 + 11 + 8 = 35 marks]

5 In a certain country, it has been estimated that 8% of the population has a rare blood disease (A). When tested, 95% of those with the disease tested positive (+). $2\frac{1}{2}$% of those that did not have the disease also tested positive.

a Complete the tree diagram below.



b What is the most likely scenario?

c A person from those who had had the blood test was randomly selected.

Determine the following probabilities

i $P(\text{The person had the disease and tested positive})$

ii $P(\text{The person tested positive})$

iii $P(\text{The test will be incorrect})$

iv $P(\text{The person did not have the disease} \mid \text{he tested positive})$

d If 400 people tested positive, how many of these would you expect to not have the disease? Comment on this result.

[5 + 1 + 5 + 3 = 14 marks]

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